

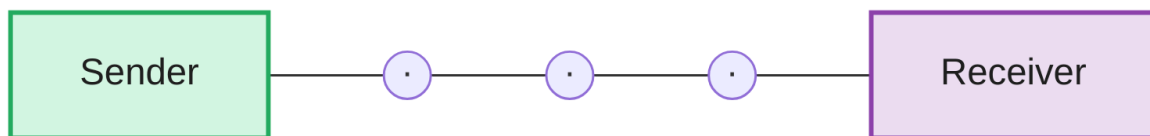
Lecture 4: IP Addressing in Depth

IPv4 and IPv6 use different bit-lengths because they need to support very different numbers of devices. IP address classes follow clear binary patterns, and addresses are distributed worldwide through a simple hierarchy — from global authorities down to individual ISPs.

Why is IPv4 32-bit?

A **standard** was fixed at 32 bits so that every device reads and interprets an IP address the same way — without this fixed length, systems could get confused about where one address ends and another begins.

IPv4 → 32 bits → 2^{32} devices ≈ 400 crore (4.3 billion) devices



Why is IPv6 128-bit?

IPv4's 32 bits were **extended** to 128 bits:

$$32 \times 4 = 128$$

IPv6 → 128 bits → 2^{128} devices (planned to be sufficient for ~200 years into the future)

Why so large? Because the number of internet-connected devices keeps growing rapidly, and IPv6 was designed with enough extra space that address exhaustion won't be a problem again for a very long time.

IPv6 Format

IPv6 addresses are written in **hexadecimal**, grouped into 8 blocks of 4 hex digits each, separated by colons:

```
Minimum: 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000
Maximum: FFFF : FFFF : FFFF : FFFF : FFFF : FFFF : FFFF : FFFF
```

Why 4 hex digits per block?

```
1 hex digit = 4 bits
F (15) is the largest digit in hexadecimal → needs 4 bits to represent ( $2^4 = 16$ 
values: 0-15)

4 bits × 4 digits = 16 bits per block
16 bits × 8 blocks = 128 bits total
```

*This hexadecimal format is for **human readability**. Internally, the computer still reads and processes the address in pure **binary**.*

Why Wasn't IPv5 Used?

There was a version called IPv5, but it was **never implemented as a public addressing standard** because it was not designed to be sufficient for future scaling needs — so the world moved from IPv4 directly to IPv6.

Binary Structure of IP Address Classes

Each class is identified by a fixed pattern of **leading bits** in the first octet (8 bits). The more bits that are fixed, the fewer bits remain free — and those free bits decide how many addresses (and what range) that class gets.

Class A

```
First bit fixed to: 0
Remaining 7 bits free
```

```
0 00000000 → 00000000
0 11111111 → 01111111
```

```
 $2^7 = 128$  possible values → Range: 0 - 127
```

Class B

First 2 bits fixed to: 1 0
Remaining 6 bits free

10 000000 = 128
10 111111 = 191

$2^6 = 64$ possible values → Range: 128 – 191

Class C

First 3 bits fixed to: 1 1 0
Remaining 5 bits free

110 00000 = 192
110 11111 = 223

$2^5 = 32$ possible values → Range: 192 – 223

Class D

First 4 bits fixed to: 1 1 1 0
Remaining 4 bits free

1110 0000 = 224
1110 1111 = 239

$2^4 = 16$ possible values → Range: 224 – 239

Class E

First 4 bits fixed to: 1 1 1 1
Remaining 4 bits free

1111 0000 = 240
1111 1111 = 255

$2^4 = 16$ possible values → Range: 240 – 255

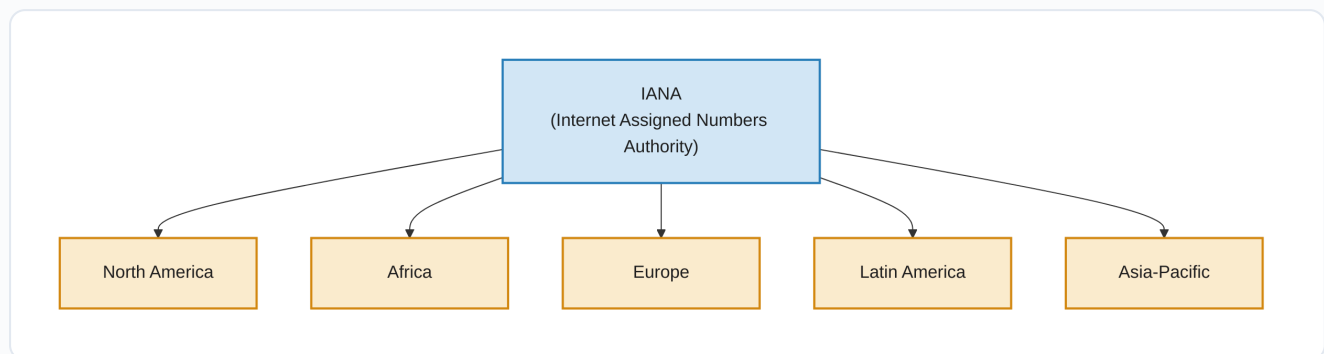
Summary of Binary Patterns

CLASS	FIXED LEADING BITS	RANGE
A	0	0 – 127
B	10	128 – 191
C	110	192 – 223
D	1110	224 – 239
E	1111	240 – 255

Who Distributes IP Addresses Globally?

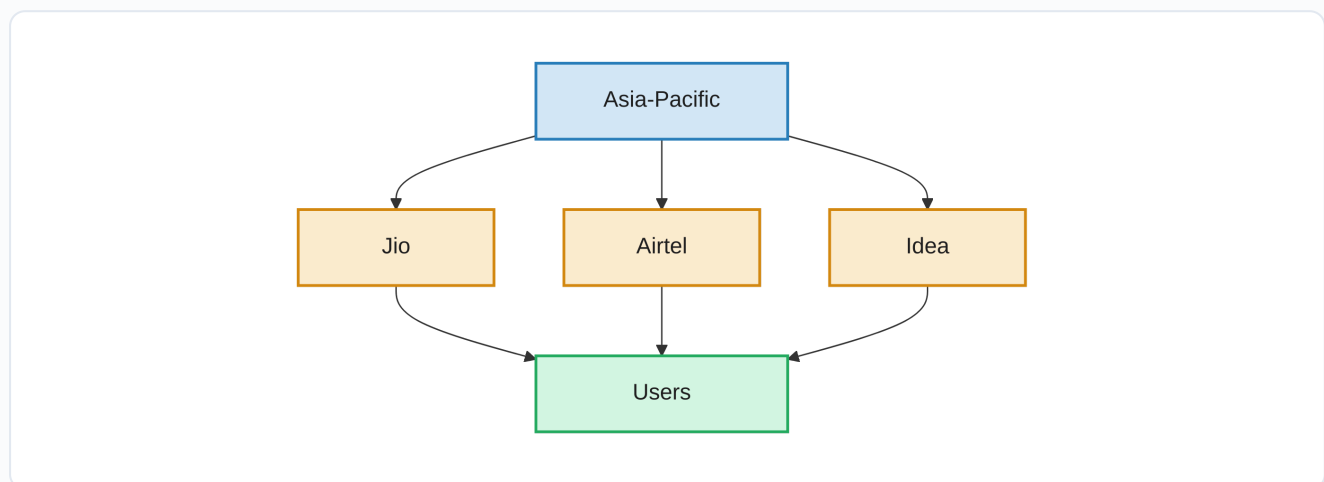
IANA — Internet Assigned Numbers Authority is the body that distributes IP addresses worldwide.

IANA divides the world into **5 regions**:



Each region then distributes IP addresses **country-wise to ISPs**.

Example: Asia-Pacific → India



Why were addresses divided into classes?

Different ISPs need very different **quantities** of IP addresses:

- ♦ Some ISPs need **crores (tens of millions)** of addresses — but such large ISPs are **few** in number.
- ♦ Some need **lakhs (hundreds of thousands)** — a **larger** number of such ISPs exist.
- ♦ Some need only **thousands** — and this is the **most common** requirement.

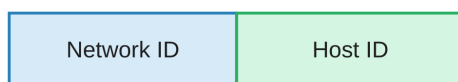
*Classes exist so that IP addresses can be allocated in **appropriately sized blocks** matching each ISP's actual need, instead of wasting large blocks on small ISPs.*

Illustrative example of demand:

REQUESTER	ADDRESSES NEEDED
Idea	2 crore
Jio	5 crore
Airtel	4 crore
ISP X	5 lakh
ISP Y	50,000
ISP Z	10,000

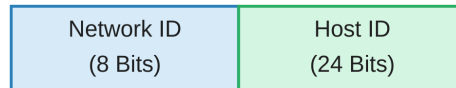
Network ID and Host ID

Every IP address in a class is split into two parts:



- ♦ **Network ID** identifies which network a device belongs to (this part is given to the ISP).
- ♦ **Host ID** identifies the specific device within that network.

Class A



Network ID $\rightarrow 2^7 = 128$ networks (first bit fixed as 0)

Host ID $\rightarrow 2^{24} = 1.67$ crore (≈ 16.7 million) addresses per network

Reserved Addresses in a Network Block

Each network has 2^{24} possible Host IDs. Two of these are reserved (the Network Address and the Broadcast Address), leaving $2^{24} - 2$ usable addresses for devices.

RESERVED ADDRESS	PURPOSE
Network Address	Identifies the network itself — assigned when a device first requests an IP
Broadcast Address	Used to deliver a message to all devices on that network at once

Network Address — the address a user's device effectively starts from when it requests an IP for the first time.

Broadcast Address — used when a message needs to reach every device in the network.

Example:

73.0.0.0 $\rightarrow$ Network Address
73.255.255.255 $\rightarrow$ Broadcast Address

Everything **between** the network address and broadcast address is available for **public use** (actual devices).

Special Reserved Values in Class A

0 $\rightarrow$ Reserved (0.0.0.0 is used as a default placeholder IP before the ISP allocates a real one)

127 $\rightarrow$ Reserved as the Loopback Address (used by a device to refer to itself)

Usable range for the public: 1 - 126
 $2^7 - 2 = 126$ usable network blocks

Class B

Network ID (16 Bits)	Host ID (16 Bits)
-------------------------	----------------------

Network ID → first 2 bits fixed as "10", 14 bits free → $2^{14} \approx 16,000$ networks
Host ID → $2^{16} \approx 65,000$ addresses per network ($2^{16} - 2$ usable, after
reserving
network & broadcast addresses)

Example allocation scenario:

If an ISP needs 1 lakh (100,000) addresses, a single Class B network (~65K) isn't enough — so **2 networks** would be allocated:

Network 1 → ~65K IP addresses
Network 2 → ~65K IP addresses

Class C

Network ID (24 Bits)	Host ID (8 Bits)
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Network ID → first 3 bits fixed as "110", 21 bits free → 2^{21} networks
Host ID → $2^8 - 2 = 254$ usable addresses per network

Class C networks are much smaller (only 254 usable hosts each) — suited to ISPs or organizations needing only a small block of addresses.

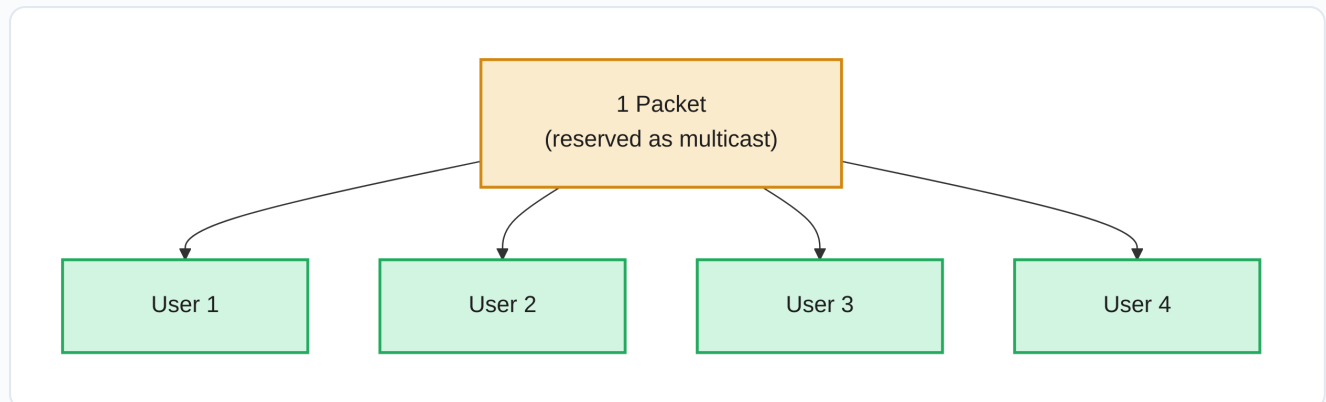
Class D — Reserved for Multicasting

First 4 bits fixed as: 1 1 1 0

Remaining bits → 2^{28} addresses

Multicasting means sending the **same single packet** to a specific group of multiple recipients — without sending an individual copy to each one.

Example: Zoom Call



- ✦ A single packet is marked for multicasting and delivered to **all participants at once**.
- ✦ Without multicasting, the sender would need to send a **separate packet to every single user individually** — which does not scale, and could otherwise crash the system under heavy load.

Packet structure for multicasting:



Here, the **Multicasting Address** takes the place of a single destination IP address, since the packet is meant for a *group*, not one device.

Class E — Reserved for Future Use

First 4 bits fixed as: 1 1 1 1

Remaining bits → 2^{28} addresses

Class E is reserved by **IANA** for **future/experimental use** and is not used for general public or commercial allocation.

Broadcasting vs Multicasting

BROADCASTING	MULTICASTING
Delivers to all devices within the same network	Can deliver to devices across different networks too
Used for network-wide announcements	Used for group communication (e.g., video calls, streaming to a subscriber group)

Key Points on Class Distribution

- ✦ IANA distributed all classes of IP addresses across the **5 world regions**, which further distribute them **country-wise to ISPs**.
- ✦ Within all classes, certain **private IP ranges** are also reserved (for internal/local network use, not publicly routable).

Summary

- ♦ IPv4 uses a fixed 32-bit standard so every device reads addresses consistently, giving about 2^{32} (~400 crore) possible addresses.
- ♦ IPv6 extends this to 128 bits (32×4), giving 2^{128} addresses — written in hexadecimal (8 blocks of 4 hex digits, 16 bits each) for human readability, though computers still process it in binary.
- ♦ IPv5 was never implemented as a public standard because it wasn't built to scale for future needs.
- ♦ Each IP address class is defined by a fixed pattern of leading bits: Class A starts with 0, Class B with 10, Class C with 110, Class D with 1110, and Class E with 1111 — the number of remaining free bits determines that class's address range.
- ♦ IANA (Internet Assigned Numbers Authority) distributes IP addresses globally across 5 regions (North America, Africa, Europe, Latin America, Asia-Pacific), which then allocate addresses country-wise to ISPs like Jio, Airtel, and Idea.
- ♦ Classes exist because different ISPs require very different quantities of addresses — some need crores, others lakhs or thousands — so classes allow appropriately sized blocks to be allocated.
- ♦ Every IP address splits into a Network ID (identifies the network) and a Host ID (identifies the device); Class A uses 8-bit Network ID / 24-bit Host ID, Class B uses 16/16, and Class C uses 24/8.
- ♦ In every network block, the first address (Network Address) and the last address (Broadcast Address) are reserved, leaving $2^n - 2$ usable addresses for devices.
- ♦ In Class A specifically, 0 is reserved as a default placeholder IP and 127 is reserved as the Loopback Address, leaving 1–126 for public use.
- ♦ Class D is reserved for multicasting — sending one packet to a defined group of recipients at once (e.g. a Zoom call), avoiding the need to send individual packets to every user.
- ♦ Class E is reserved by IANA for future and experimental use.
- ♦ Broadcasting delivers a message to all devices within the same network, while multicasting can reach a defined group of devices even across different networks.